# *Business Plan Section - WRO*

## 4.4 The problem, in numbers

Mohenjo-daro covers 240 hectares, of which about one third has been excavated. The people responsible for maintaining it are insufficient, as shown:

| Measure | Number | Source |
|---|---|---|
| Labourers required | 350 | Sunday Telegraph, 2013 |
| Labourers present on the day of the visit | 16 | Sunday Telegraph, 2013 |
| Curators and conservators on site | 2 | Herald, 2017 |

Two conservators cannot inspect 240 hectares on any useful schedule, and even if they could, human inspection produces no comparable record. A conservator can see that a wall looks damp. He cannot see that it is two degrees cooler than it was in March, or that the salt crust has grown four percent since the last visit. Different person, different day, different light, different opinion, and no dataset at the end of it.

## 4.5 What we sell

We sell an instrument, not a service. The customer buys a rover, we install it on their wall, and we train their existing staff to operate it. After that it belongs to them and they run it without us.

**What the customer receives:** the rover, assembled and calibrated; the dock and guide markers, installed; firmware, at no separate charge; two days of staff training and a printed manual; a spares kit. The rover writes ordinary files rather than an application the site must maintain: a CSV of readings, a PDF report, and a heatmap image of the wall. These open in Excel or print on paper, need no internet connection, and will still open in twenty years.

## 4.6 Why a sale and not a service

| Reason | Detail |
|---|---|
| Travel cost | Mohenjo-daro is 500 km from Karachi. A recurring service would spend most of its revenue on travel rather than on conservation. |
| Procurement | Government departments approve one-time capital purchases far more readily than recurring charges, which must survive every budget round. |
| Succession | We are students and will go to university. A system only we can operate dies when we leave. A system the site operates outlives us. |

This is why the training line in our costing is explicit rather than absorbed elsewhere. The independence of the site is the model, so the training has to be real and budgeted.

## 4.7 Who buys it

| Customer | Site | Why they fit |
|---|---|---|
| Department of Archaeology and Antiquities, Sindh | Mohenjo-daro, 240 ha | Documented saline action from a rising water table; mud capping already the standard treatment; two conservators for the whole site |
| Same department | Makli Necropolis, Thatta | Also a Pakistani World Heritage Site with salt and monsoon damage. UNESCO sent an emergency mission there alongside Mohenjo-daro in 2022. |
| Federal Department of Archaeology | Taxila, Rohtas Fort, Takht-i-Bahi | Other Pakistani World Heritage Sites with exposed masonry |
| Ministry of Culture, Peru | Chan Chan | UNESCO lists lack of monitoring system, salt contamination of the mud-brick and rising ground water among the factors affecting the property. On the danger list since 1986. |
| UNESCO and heritage foundations | Any of the above | Fund conservation and need evidence of outcomes |

| | | |
|---|---|---|
| Museums and private heritage trusts | Historic buildings with rising damp | Smaller decisions, faster, often better funded |

Our first target customer is the **Department of Archaeology and Antiquities, Sindh.**

## 4.8 What the instrument gives them

The rover measures the same wall patches repeatedly, at the same distance, with a calibration check every patrol. That is what makes a reading in March comparable with one in September.

| Benefit | What it means in practice |
|---|---|
| Timing | Knowing when a wall started wicking, or when a mud cap has saturated and stopped working |
| Measurement | A record of how a wall behaves through a year, which is not published anywhere for this site |
| Evidence | Support for funding applications, where money at this site moves in steps of Rs 1 million and Rs 30 million |

## 4.9 Cost and price

| Costs | PKR | USD |
|---|---|---|
| Sensing and vision | 52,000 | 186 |
| Compute | 27,000 | 96 |
| Mechanical and enclosure | 18,000 | 64 |
| Power system | 12,000 | 43 |

| | | |
|---|---|---|
| Drivetrain | 9,000 | 32 |
| Control electronics | 6,000 | 21 |
| Navigation | 4,000 | 14 |
| **Hardware subtotal** | **128,000** | **457** |
| Assembly, testing and calibration | 32,000 | 114 |
| Teaching and training, two days | 20,000 | 71 |
| **Delivered cost per unit** | **180,000** | **643** |
| **Sale price** | **240,000** | **857** |
| **Surplus per unit** | **60,000** | **214** |
| of which held as warranty reserve | 25,000 | 89 |
| **Net contribution to the next build** | **35,000** | **125** |

*Hardware priced at a build quantity of ten units, from our own bill of materials using Karachi supplier prices. USD indicative at 280 PKR per dollar, August 2026.*

Sensing and vision is the largest hardware line at 40% of the total, which is why the thermal sensor decision in section 3.8 matters as much to the cost model as to performance.

**Installation and travel are covered by the host organisation** and are not in our delivered cost. We install and train on site, and the customer provides local transport and accommodation, which is standard practice for a visiting technical team delivering equipment a department has requested.

Our fixed costs are zero: school workshop, our own time, self-made firmware. The project therefore breaks even on the first unit sold.

## 4.10 Revenue streams

| Stream | Price | Who it suits | Status |
|---|---|---|---|
| Unit sale | PKR 240,000 (USD 857) | Departments with a capital budget or grant funding behind them | Primary stream |

| | | | |
|---|---|---|---|
| Return-to-base repair | PKR 15,000 labour, plus freight both ways paid by the customer, plus cost of parts is separate. | Faults trained site staff cannot resolve from the spares kit | Small but recurring |
| Consumables and spare parts | Cost plus 10% surplus | Any site in operation | Small but recurring |

The first six months are covered by warranty, funded from the PKR 25,000 reserve. After that the site ships the rover or the affected module to Karachi and pays courier both ways. We do not price repairs to earn from them: our model depends on the site operating independently, and a business that earns more when its product fails has the wrong incentive.

We deliberately do not charge for an annual subscription, a service contract etc. Additionally, Training and firmware are included in the purchase.

## 4.11 Market entry and early funding

The **first unit** is offered at cost, PKR 180,000, or **free if grant funding** can be secured, in exchange for site access, permission to publish the data, and a reference. We are asking an institution to be first, which carries all of the risk and none of the precedent. The PKR 60,000 forgone buys the one thing we cannot manufacture: a working deployment at a named World Heritage Site. Nobody buys the second unit from a team with no installations. **This is a customer acquisition cost**

At under 900 USD the instrument is within reach of international heritage funding, where a single ground-penetrating radar consultant survey producing one snapshot costs several thousand dollars.

Before any sale, the project is funded by competition prizes, school support and sponsorship. This is start-up funding rather than revenue: it gets us to a first working deployment, after which unit sales sustain the project. Because our fixed costs are zero, the funding requirement is the cost of building units, not of running an organisation.

## 4.12 Business model canvas

| Block | Content |
|---|---|
| Key partners | Department of Archaeology and Antiquities, Sindh; UNESCO Pakistan; Karachi universities; local electronics suppliers |

| | |
|---|---|
| Key activities | Building and calibrating rovers; installation and commissioning; training site staff |
| Key resources | The calibration method that makes readings comparable between visits; site access and relationships; the training curriculum |
| Value proposition | An affordable instrument that measures how a heritage wall changes over time, operated by the site's own staff |
| Customer relationships | One-time sale plus training, after which the site is independent |
| Channels | Direct to provincial archaeology departments; UNESCO and heritage funders; conservation conferences |
| Customer segments | See the customer table in 4.7 |
| Cost structure | Hardware 128,000; assembly 32,000; training 20,000; delivered cost 180,000; installation hosted by customer; fixed costs zero |
| Revenue streams | Unit sale at PKR 240,000; return-to-base repair at PKR 15,000 plus parts; consumables at cost plus 10% |